

A Appendix

A.1 Complementary details of the reviewed works

Table 4: Models grouped by modality.

Modality	Models
Text	BART [39], LED [59], BigBird [60], LongT5 [40], Word2Vec [80], GPT-2 [36], OPT [55], LLaMA 2 [37], GPT-Neo [76], T5 [40], Vicuna [83], Alpaca [64], T5 Flan [54], GPT-2-XL [36], DeBERTa [66], RoBERTa [67], Pythia [68], LLaMA 3 [37], Gemma [69], Baichuan2 [70], DeepSeek-R1 [71], GLM [72], Qwen2.5 [73], Mistral [74], BERT [35] (monolingual and multilingual), XLM-R, XGLM, LLaMA-3.2 [38], Phi-2 [78]
Speech	Wav2Vec2.0 [41], HuBERT [42], WavLM [56]
Image	ViT-H [43], ViT [43], ResNet-50 [91]
Video	VideoMAE [97], ViViTB [92]
Image + Text	InstructBLIP [45], mPLUG-Owl [46], IDEFICS [82], CLIP [44], BLIP-L [93], BridgeTower [88], GIT [89], LLaVA [85], FLAVA [84], Qwen2.5-VL [86].
Video + Text	LLaVA-OV-7B [94]
Video + Speech	TVLT [96]
Speech to Text	Whisper [52]
Multimodal	ImageBind [95]